

Pallav Gupta

Senior Software Engineer | Platform & Distributed Systems | Cloud-Native

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PROFESSIONAL SUMMARY

Platform and distributed-systems engineer with 7+ years building backend systems end to end, across telecom, AI, and enterprise domains. Comfortable owning systems from the first whiteboard sketch through production and everything after. Sole architect at Nokia of a real-time platform orchestrating multiple AI models at the edge, cutting API latency by 40% and scaling it to handle 4x the load per device. Turns vague business problems into systems that ship, hold up under real traffic, and stay simple enough for the next person to build on.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript (ES6+), Shell Scripting

Backend & APIs: FastAPI, Flask, Spring Boot, Node.js, Express.js, NestJS, REST, HTTP/2, JWT

Distributed & Event-Driven: Microservices, Event-Driven Architecture, MQTT, ZeroMQ, Message Queues, Distributed Systems, Edge Computing, Eventual Consistency

Databases: PostgreSQL, MongoDB, ClickHouse, MariaDB, Oracle DB, ORM

AI Systems: Multi-model inference serving (NVIDIA Triton), real-time inference over APIs, model orchestration

Cloud & DevOps: Docker, Kubernetes, K3S, Helm, ArgoCD, GitOps, GitLab CI/CD, Microsoft Azure, Nginx, Apache2

Practices: System Design, Performance Optimization, SOLID, Domain Modeling, Testing, Observability

PROFESSIONAL EXPERIENCE

HCL Software | Bengaluru, India Apr 2026 – Present

Senior Software Engineer 2 · *Augmented Network Automation*

Stack: Python, FastAPI, ClickHouse, MQTT, ZeroMQ, Docker, Helm, Kubernetes (K3S), HTTP/2

- Ran the Bitbucket-to-GitLab migration for 6 teams and 75+ engineers, planning the cutover so CI/CD pipelines and release cadence kept running with zero downtime.
- Migrated the persistence layer from ClickHouse to MongoDB and introduced a new ORM, which gave us more flexible queries and a data model that stayed maintainable as the product grew.
- Set up one-click, Helm-based deployment on Kubernetes (K3S) and cut deployment time in half, with releases that were finally repeatable across every environment.

Nokia | Bengaluru, India Apr 2022 – Apr 2026

Platform Lead · *Context-Aware AI Positioning Platform*

Stack: Python, FastAPI, PostgreSQL, MQTT, ZeroMQ, Docker, Helm, Kubernetes (K3S), HTTP/2

- Was the sole architect of a cloud-native platform that orchestrates multiple AI models for real-time context-aware positioning, and carried it from the first whiteboard sketch all the way to production.
- Added multi-model inference on NVIDIA Triton, served over an API, so the team could onboard any number of AI models and get them deployed 60% faster.
- Designed the event-driven backbone over MQTT and ZeroMQ, leaning on async messaging for the low-latency communication the distributed edge components needed.
- Profiled and tuned the API, messaging, and deployment layers to cut API latency by 40% and support 4x more camera streams per edge device, which improved the deployment economics for customers.
- Built the containerized deployment workflows with Docker, Helm, and K3S that kept edge and IoT nodes highly available across the fleet.

Platform Lead · *Application Management Platform*

Stack: Node.js, Express.js, Spring Boot, Python, ReactJS, PostgreSQL, MQTT, ArgoCD, Kubernetes, Helm

- Led the build of a self-service developer platform where teams could upload, deploy, and run their own applications at enterprise scale without waiting on a central ops team.
- Wrote the backend across Node.js, Spring Boot, and Python and built a ReactJS operator console, so teams had one place to manage everything.
- Added a security check that scanned every uploaded container image and blocked malicious ones before they could run.
- Surfaced real-time health, telemetry, and metrics over MQTT so teams could see what was actually happening across distributed nodes.
- Replaced hardcoded setup with a dynamic, GitOps-driven pipeline on Kubernetes, Helm, and ArgoCD, which brought application onboarding time down by 90%.

Infosys | Bengaluru, India Dec 2021 – Apr 2022

Associate Consultant · *Common Authentication System*

Stack: ReactJS, Node.js, Express.js, NestJS, MongoDB, Apache, Shell Scripting

- Built a centralized authentication platform as a reusable plugin, so new enterprise apps could drop it in instead of rebuilding auth every time.
- Implemented token-based authentication and authorization with JWT, giving every integrating service a standard way to verify users and enforce access.
- Migrated core backend services from Express.js to NestJS, which tidied up the structure and made the codebase easier to maintain.
- Configured Apache for routing, load balancing, and SSL termination across environments.

Tata Consultancy Services | Gandhinagar, India Jul 2019 – Dec 2021

Systems Engineer · *Telecom Solutions & Cloud Infrastructure*

Stack: Node.js, ReactJS, Spring Boot, Azure, PostgreSQL, MariaDB, Oracle DB, Shell Scripting

- Built and maintained backend services and APIs for telecom customers across India and Europe using Java Spring Boot and Node.js.
- Built reusable ReactJS component libraries that several teams shared, which sped up frontend work across products.
- Owned deployment, scaling, and operational support for application and database infrastructure on Azure, spanning PostgreSQL, MariaDB, and Oracle DB.
- Set up secure ExpressRoute VPN connectivity between Azure services and customer on-premises environments.

EDUCATION

Madhav Institute of Technology and Science | Gwalior, India 2015 – 2019

Bachelor of Engineering, Mechanical Engineering | CGPA 7.1

Made the move into software engineering through self-study alongside work, and joined TCS as a Systems Engineer after graduating in 2019.